

Yuetong Li

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EDUCATIONAL BACKGROUND

ShanghaiTech University	Shanghai, China
School of Information Science and Technology	Sep.2023-Jun.2027(expected)
Major in Computer Science and Technology	
Cumulative GPA: 3.88/4.0 Rank: 4/170	
Selected Courses: Discrete Mathematics, Intro to Programming, Algorithms and Data Structures, Probability and Statistics, AI in Medical Imaging, Machine Learning, Computer Architecture, Control Theory, Computer Graphics, Mechatronics, Artificial Intelligence, Intro to Robotics, Digital Signal Processing	

RESEARCH EXPERIENCE

AURORA: Active Uncertainty-Driven Re-Orientation for In-Hand 3D Reconstruction	Feb.2026-Present
Supervised by Prof. Chenxi Xiao ShanghaiTech Robot Interaction and Manipulation Lab	
Objective: Propose an uncertainty-aware closed-loop planning framework for in-hand object reconstruction, actively selecting reorientation actions to expose under-observed surfaces.	
Core Contents:	
- Designed Ray-GPIS-based geometric uncertainty estimation, using Gaussian Process Implicit Surfaces to quantify reconstruction confidence from sparse multi-view point clouds. - Developed a Next-Best-View (NBV) planner that maps uncertainty gradients to feasible in-hand rotation primitives via Lie algebra-based pose reasoning. - Implemented a robust online fusion pipeline combining RGB-D perception, keyframe filtering, object-centric point cloud fusion, and uncertainty-aware replanning. - Demonstrated faster and more complete reconstruction than open-loop and single-view baselines, achieving higher Mesh-Mesh F-scores within limited interaction budgets on real robotic hardware.	

AWARDS & HONORS

Merit Student ShanghaiTech University	Dec.2025
Second Prize National Electronics Design Contest Shanghai Region	Oct.2025
First Prize ShanghaiTech Electronics Design Contest	May 2025
Merit Student ShanghaiTech University	Dec.2024

PROJECT EXPERIENCE

Multi-Agent Coordination in Overcooked via Macro-Expectimax and Neural Q-Learning	2026 Spring Semester
- Extended expectimax from primitive actions to task-level macro-actions, allowing shallow search to reason over long task sequences; implemented Dijkstra-based executors for navigation and interaction. - Developed a Bayesian teammate-intent model that updates beliefs over the partner's latent macro-action from observed behavior and incorporates them into partner-aware expectimax planning. - Built a CNN-based approximate Q-learning agent with parameter-shared self-play, Double-Q-style targets, prioritized replay, reward shaping, and multi-step returns for sparse-reward coordination. - Evaluated both approaches across 38 layouts; macro-action planning matched or exceeded Greedy on 25 layouts and achieved its strongest gains in coordination-intensive and deadlock-prone scenarios.	
Vision-Based Distance-Constrained Tracking and 3D Reconstruction of Unknown Objects	2025 Fall Semester
- Developed a closed-loop embedded perception-decision-execution system using FSM control logic and UART communication for reliable low-latency module interaction. - Processed Intel RealSense RGB-D data with SAM2 for target segmentation and integrated BundleTrack with SIFT, RANSAC, and graph-based matching for accurate 6D pose tracking. - Built a 3D reconstruction pipeline by aligning multi-view point clouds into a unified coordinate system and applying denoising, downsampling, and TSDF fusion for high-fidelity mesh reconstruction.	
Real-time Global Illumination via Radiance Cascades (2D/3D Implementation Study)	2025 Fall Semester
- Implemented a Radiance Cascades based global illumination system in OpenGL/GLSL, using multi-scale probe cascades, directional radiance sampling, and hierarchical cascade merging for efficient indirect light propagation. - Designed a interactive 2D real-time rendering pipeline with SDF(Signed Distance Fields)-guided probe tracing, ping-pong FBO based SDF updates, hierarchical cascade merging, and angular integration, enabling brush-based light painting/erasing at 60+ FPS at 1920*1080 resolution.	

- Extended the method to static 3D scenes by mapping surface probes into a high-resolution radiance atlas, combining offline ray-traced baking with hierarchical cascade sampling, and enabling real-time atlas-based rendering with dynamic camera interaction.

Motion-Sensing Gesture Mouse Design Based on the MPU6050 and Flex Sensors 2025 Fall Semester

- Designed and implemented a wearable gesture-controlled mouse system using STM32, MPU6050 IMU, and flex sensors for real-time cursor and gesture interaction.
- Developed sensor fusion and motion tracking algorithms using I2C communication and complementary filtering to combine gyroscope and accelerometer data for stable hand pose estimation.
- Implemented real-time mouse control via the HID Mouse protocol, supporting cursor movement, clicking, and scrolling through hand gestures and finger bending actions.

ML-Patch: Evaluating Hidden Knowledge in LLMs via Multi-Layer Patching 2025 Spring Semester

- Proposed ML-Patch, a novel interpretability framework that assesses factual knowledge embedded in large language models (LLMs) by patching internal hidden states across multiple layers;
- Conducted extensive experiments on 1396 Wikidata factual triples and five LLMs (e.g., GPT-J-6B, Mistral-7B, LLaMA2-13B), showing that ML-Patch consistently outperforms single-layer patching methods in knowledge attribution in prompt-based evaluation;
- Demonstrated that multi-layer interventions significantly improve the localization and recall of factual knowledge, especially in prompt-sensitive and ambiguous cases.

Performance Evaluation of Bandit Learning Algorithm 2024 Fall Semester

- Implemented and benchmarked classical multi-armed bandit algorithms (Epsilon-Greedy, UCB, Thompson Sampling) under both independent and conditionally dependent reward settings;
- Conducted over 200 simulation trials to analyze algorithmic performance against oracle-optimal policies, demonstrating Thompson Sampling's superior performance with minimal regret for dependent case;
- Implemented an intuitive Bayesian multi-armed bandit algorithms and proposed approximate dynamic programming solutions for optimization. Methods using posterior means outperformed Thompson sampling in biased cases. Introduce softmax probabilities to replace ϵ -greedy action selection and introduce gradient ascent to learn action preference.

Deep learning Cardiac Cine MRI Segmentation 2024 Spring Semester

- Developed and trained a U-Net model in PyTorch to segment the Right Ventricle, Myocardium, and Left Ventricle from cardiac Cine MRI scans;
- Evaluated critical model components (skip connections), loss functions (Dice vs. Cross-Entropy), and data augmentation, demonstrating skip connections' necessity (5.2% performance drop without them) and Dice loss's superiority for multi-class segmentation;
- Achieved mean Dice scores and compared UNet++ for enhanced feature capture, concluding optimal approaches for medical image segmentation.

Raspberry Pi-Based Handwritten Digit Recognition System 2023 Fall Semester

- Developed an integrated hardware/software solution using Raspberry Pi, camera module, and 8-segment display to capture, process, and output handwritten digits;
- Implemented k-NN algorithm with OpenCV image processing (binarization, dilation, noise filtering), solving challenges like thin digits and lighting noise;
- Designed a dedicated enclosure with fixed camera positioning and LED lighting to eliminate shadows/vignetting and ensure consistent image capture, reducing manual adjustments by 80%.

EXTRACURRICULAR ACTIVITIES

Member ShanghaiTech University Flag Guard Team	Sep.2023-Present
Volunteer ShanghaiTech University Sports Day	Oct.2024
Volunteer Shanghai Marathon	Nov.2023

PROFESSIONAL SKILLS

Programming Language: Python, C/C++, MATLAB

Miscellaneous: Visual Studio, VS Code, Pytorch, Keil, Logisim, $\text{\LaTeX}$